

Chapter 2 Exercises

Legrand Jones II

1

We want to demonstrate a version of extensionality, namely:

$$\frac{\Gamma, x : A \vdash f(x) \equiv g(x) : B(x)}{\Gamma \vdash f \equiv g : \prod_{(x:A)} B(x)}$$

Our derivation looks as follows (where due to difficulties with `mathpartir` we do not have space to express the η rule on the right hand side and we have a Γ context which is cut off):

$$\begin{array}{c} \Gamma \vdash f : \prod_{(x:A)} B(x) \\ \hline \Gamma \vdash f \equiv \lambda x.f(x) : \prod_{(x:A)} B(x) \quad \text{ETAREFL} \\ \hline \Gamma, x : A \vdash f(x) \equiv g(x) : B(x) \\ \hline \Gamma \vdash \lambda x.f(x) \equiv \lambda x.g(x) : \prod_{(x:A)} B(x) \quad \text{LAMBDAEQ} \\ \hline \text{TRANS} \frac{\Gamma \vdash \lambda x.f(x) \equiv \lambda x.g(x) : \prod_{(x:A)} B(x)}{\Gamma \vdash f \equiv \lambda x.g(x) : \prod_{(x:A)} B(x)} \quad \vdash \lambda x.g(x) \equiv g : \prod_{(x:A)} B(x) \\ \hline \text{TRANS} \frac{\Gamma \vdash f \equiv \lambda x.g(x) : \prod_{(x:A)} B(x) \quad \vdash \lambda x.g(x) \equiv g : \prod_{(x:A)} B(x)}{\Gamma \vdash f \equiv g : \prod_{(x:A)} B(x)} \end{array}$$

2

We want to derive the following:

$$\frac{\Gamma \vdash f : A \rightarrow B}{\Gamma \vdash f \circ id_A \equiv f : A \rightarrow B}$$

While a parallel derivation to the left identity derivation in the text was attempted first, the writer ran into some technical issues and found the following (conceptual) derivation more straightforward for this situation, using the following intermediate derivation:

$$\frac{\frac{\Gamma \vdash f : \prod_{(x:A)} B(x)}{\Gamma, x : A \vdash f(x) : B(x)} \text{Ev} \quad \Gamma \vdash y \equiv y' : A}{\Gamma \vdash f(y) \equiv f(y') : B(y)} \text{SubEq}$$

Expressing that if we evaluate on judgementally equal terms then our outputs are judgementally equal. We also have the from the β rule, and the definition of identities, that $\Gamma, x : A \vdash id(x) \equiv x : A$ (when Γ is a context where this all makes sense and recalling id is defined by $\lambda y.y$).

Combining these two derivations we obtain

$$\frac{\frac{\Gamma \vdash f : A \rightarrow B}{\Gamma, x : A \vdash f(x) : B(x)} \quad \Gamma, x : A \vdash id(x) \equiv x : A}{\Gamma, x : A \vdash f(id(x)) \equiv f(x) : B(x)}$$

We then finish the derivation in the same way as the text did the left unit law (I think this is kind of necessary, in other words we can't immediately appeal to the first exercise directly, since at some point we have to unwrap the definition of composition).

3

a

This is an exercise in weakening, as it morally seems it should be:

$$\frac{\frac{\Gamma \vdash Atype \quad \Gamma \vdash Btype}{\Gamma, x : A \vdash B} \text{WEAKEN} \quad \Gamma \vdash Btype}{\frac{\Gamma, y : B, x : A \vdash B}{\Gamma, y : B \vdash const_y := \lambda x.y : A \rightarrow B} \text{WEAKEN}} \text{LAMBDA}$$

b

Doing some unwrapping beforehand to make our lives easier we recall that $const_z \circ f$ is really representing $\lambda(\lambda y.z).\lambda f.\lambda x.(\lambda y.z)(f(x))$. By our β rule we see that $(\lambda y.z)(f(x)) \equiv z : C$ in our context, while also having in our context that $z \equiv (\lambda x.z)(x) : C$, which is a constant function at z , so by our first exercise we should get the desired result.

The above is more a derivation outline than anything, but should be rigorizable.

c

This should essentially follow the same outline as the above.